

MVC Pattern – House Rent Management System

1. Introduction

The **House Rent Management System** is developed using the **MVC (Model–View–Controller)** architecture. MVC divides the application into three separate components: **Model**, **View**, and **Controller**. Each component has a specific responsibility, making the application organized, scalable, maintainable, and easy to develop.

MVC Architecture

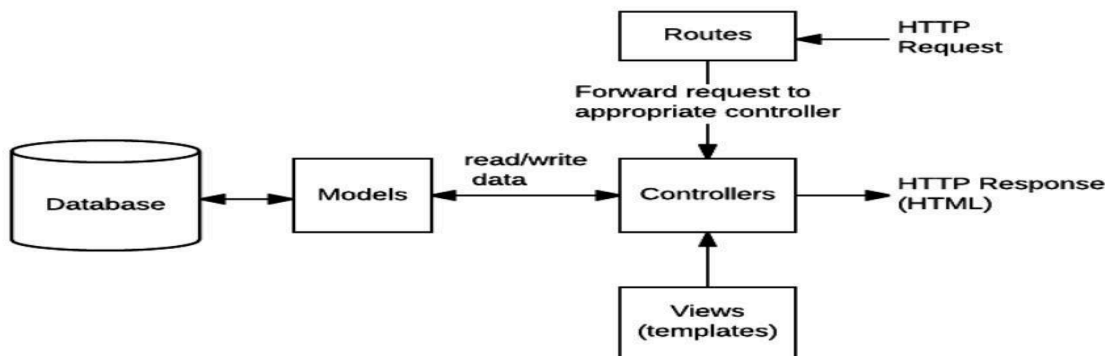
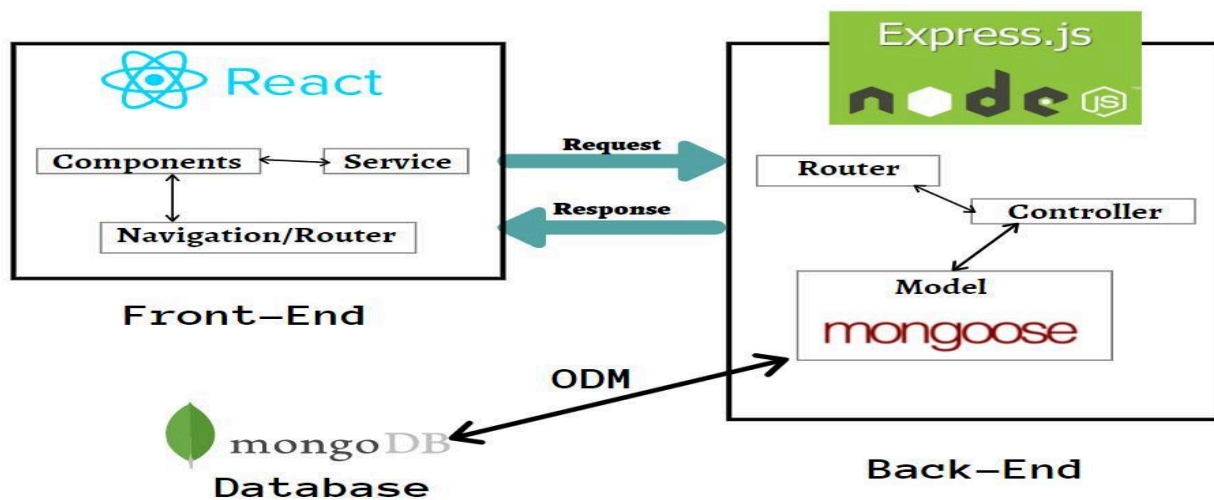
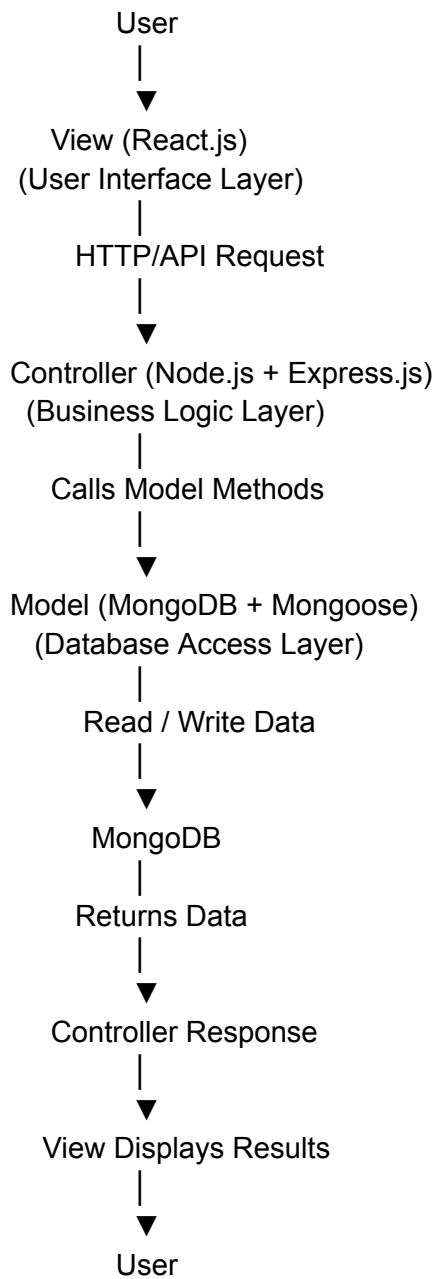
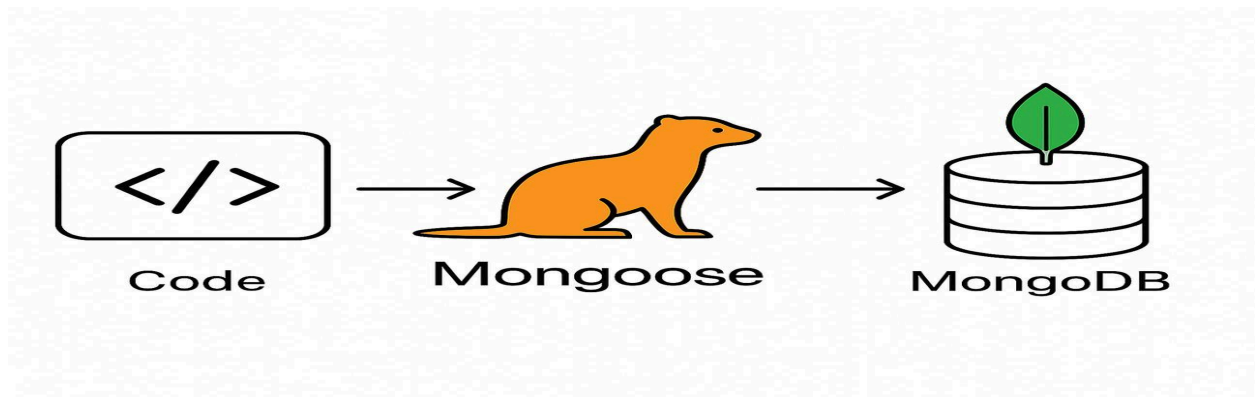


Figure 1: MVC Architecture

MVC Workflow Diagram



2. Model



Description

The **Model** represents the **data layer** of the application. It interacts with the **MongoDB** database using **Mongoose** and is responsible for storing and retrieving application data.

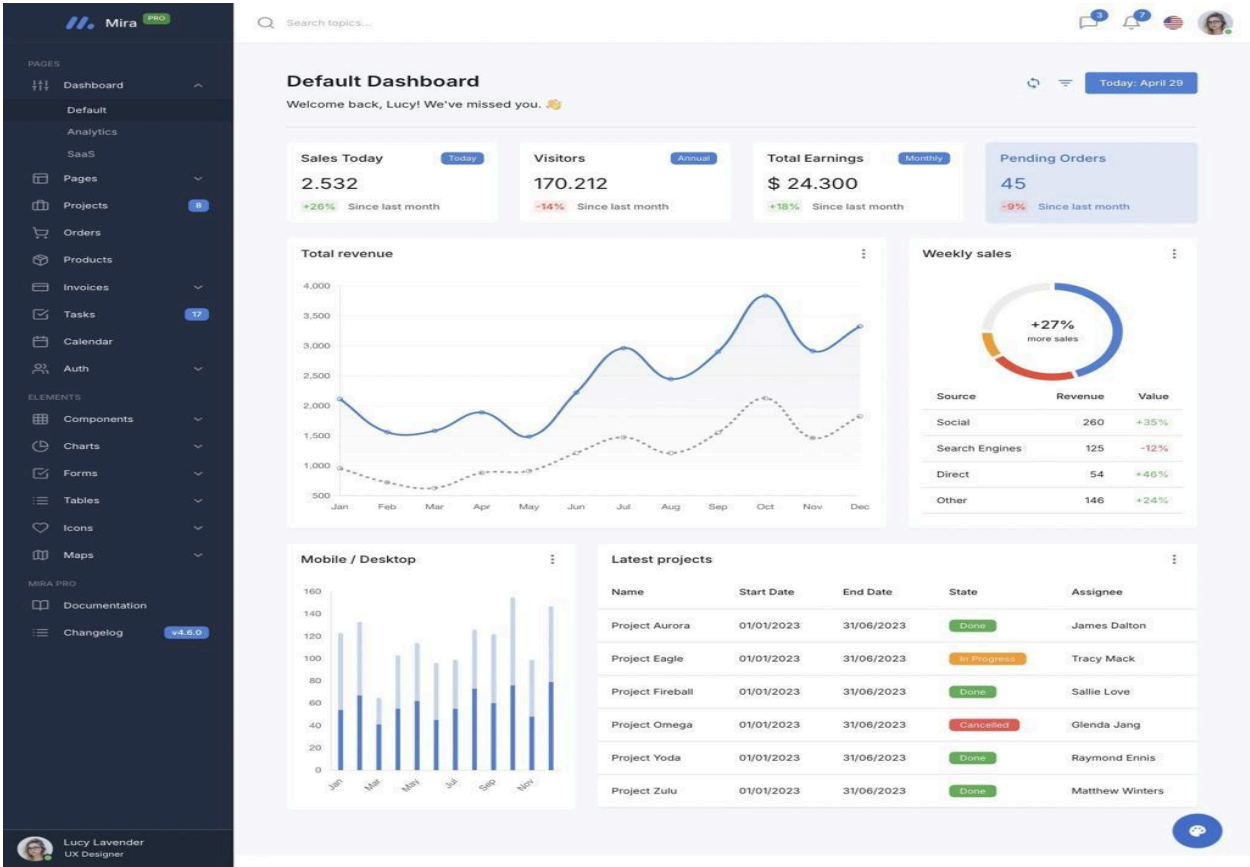
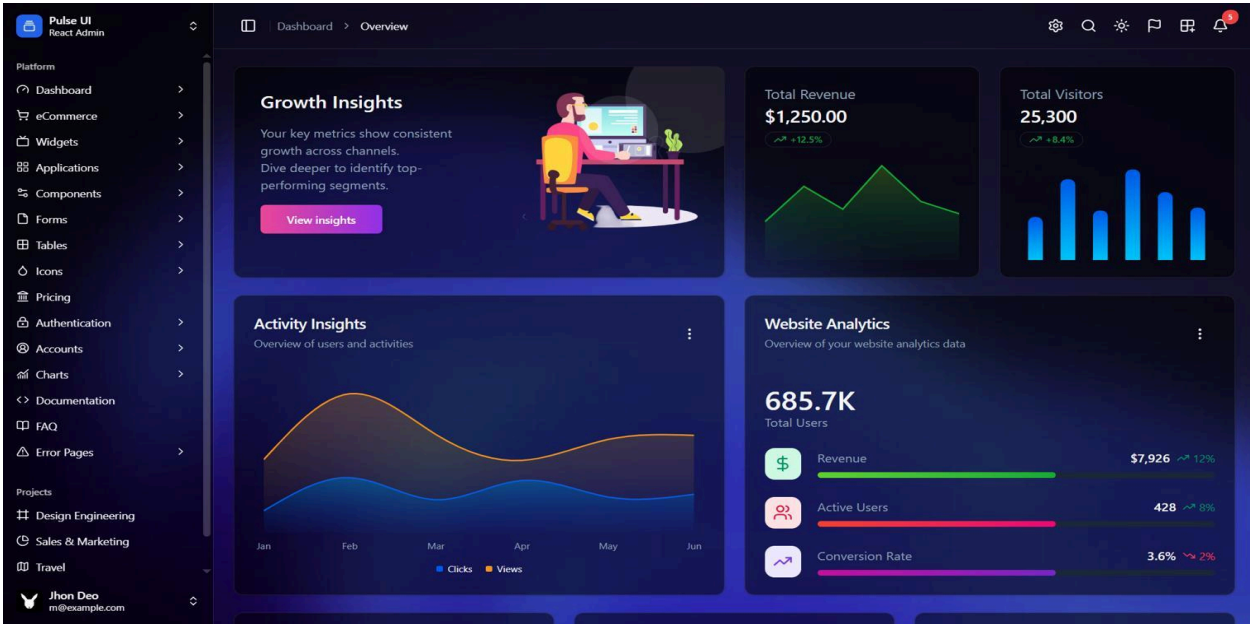
Responsibilities

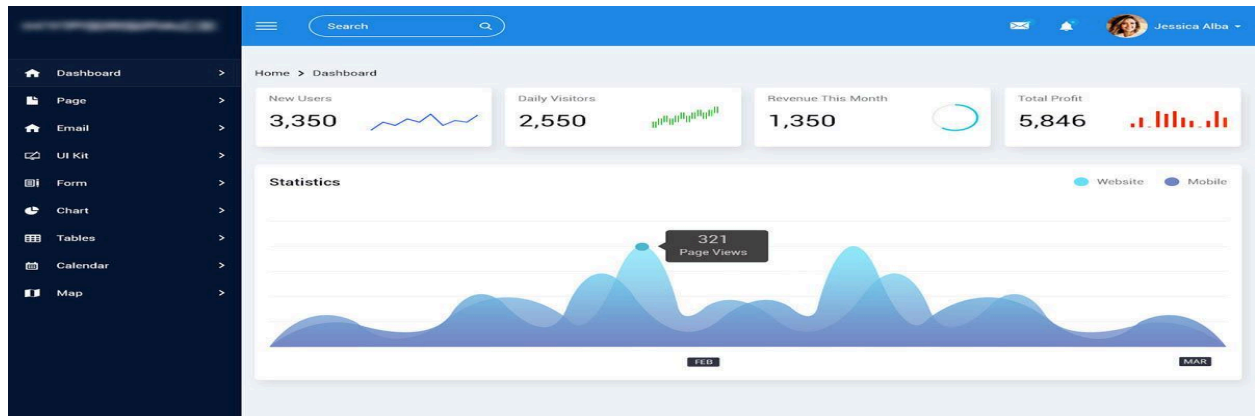
- Stores application data.
- Defines database schemas.
- Performs CRUD operations (Create, Read, Update, Delete).
- Validates data before saving.
- Communicates with MongoDB.

Models Used

- User Model
- Property Model
- Booking Model
- Review Model
- Wishlist Model

3. View





Description

The **View** is the **presentation layer** of the application. It is developed using **React.js** and provides an interactive interface for users.

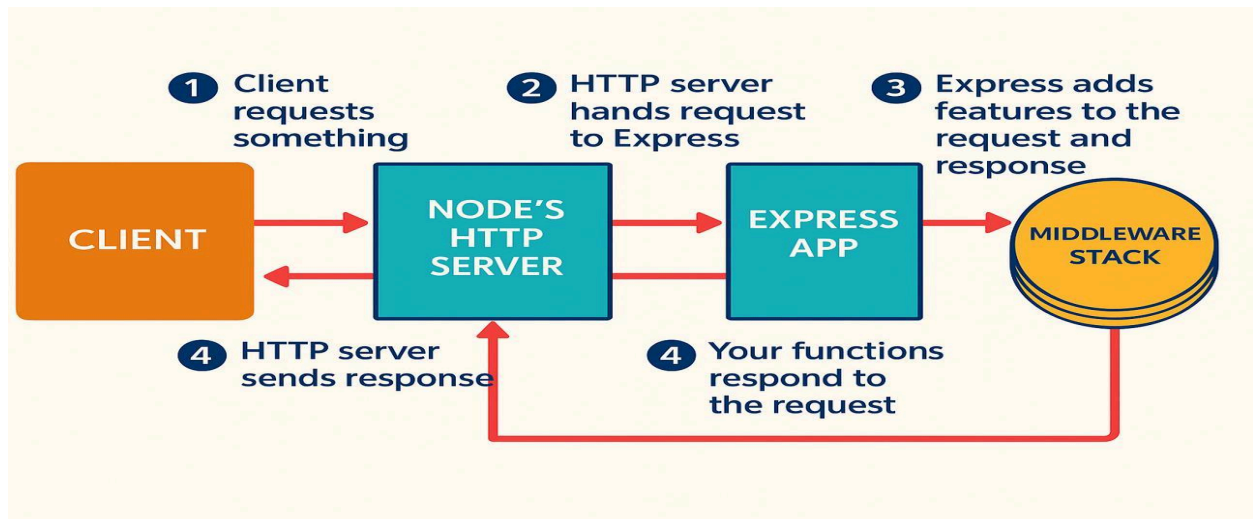
Responsibilities

- Displays web pages.
- Accepts user input.
- Sends API requests.
- Displays server responses.
- Updates the interface dynamically.

Examples

- Home Page
- Login Page
- Property Details Page
- User Dashboard
- Admin Dashboard

4. Controller



Description

The **Controller** acts as the communication layer between the **View** and the **Model**. It receives requests from the frontend, processes business logic, interacts with the Model, and sends responses back to the View.

Responsibilities

- Receives API requests.
- Validates user input.
- Processes business logic.
- Calls Model functions.
- Sends JSON responses to the frontend.

Controllers Used

- Authentication Controller
- Property Controller
- Booking Controller
- User Controller
- Admin Controller

5. MVC Request Flow

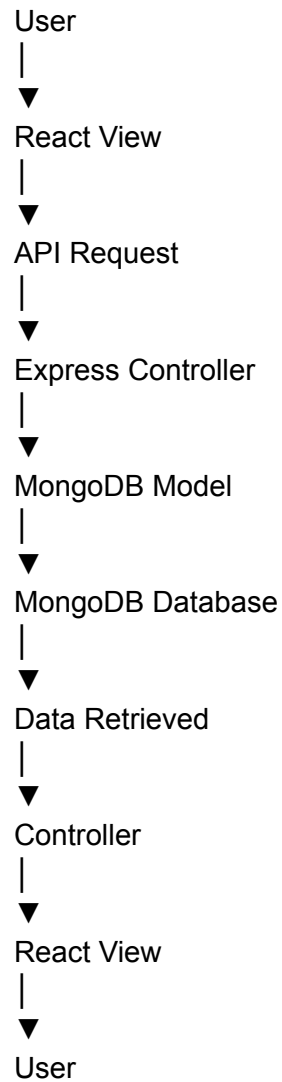


Figure 2: MVC Request Flow